

Open Mercato on Windows

One-command development environment — Setup & Troubleshooting Manual

1. What this is

A single double-click of `scripts\windows\start-windows.bat` turns a Windows machine into a complete Open Mercato development environment running in Docker: the **application** (port 3000, hot reload), the **MCP server** (port 3001), the **OpenCode AI agent** (port 4096), plus PostgreSQL, Redis, and Meilisearch. The database is created and seeded automatically. The AI assistant (`Ctrl + K` in the app) works out of the box once you provide an LLM provider API key during setup.

Every step is idempotent: re-running `start-windows.bat` is always safe, skips what is already done, and repairs what is missing.

2. Requirements

Requirement	Details
Windows	Windows 10 2004+ (build 19041, incl. 22H2/Enterprise) or Windows 11
CPU virtualization	Intel VT-x / AMD-V enabled in BIOS/UEFI (usually already on)
Memory / disk	12 GB+ RAM recommended; ~20 GB free disk
Container runtime	Docker Desktop <i>or</i> Rancher Desktop — installed automatically if missing
Administrator rights	Only for the one-time install of Docker/WSL2. Not needed afterwards, and not needed at all when IT pre-installed the runtime (see <code>-NoAdmin</code>)
LLM API key	One of: OpenAI, Anthropic, Google, Azure OpenAI / AI Foundry, OpenRouter, DeepInfra, Groq, Together, Fireworks, LiteLLM, Ollama, LM Studio

Not needed: Node.js, winget / App Installer, Visual Studio Build Tools, WSL knowledge. Everything runs inside containers; installers are downloaded directly from official sources when winget is unavailable.

3. Quick start

A. You have the repository (git clone or ZIP)

1. Double-click `scripts\windows\start-windows.bat` in the repository.
2. If Docker/WSL2 must be installed, approve the UAC (administrator) prompt. If Windows asks to restart — restart; **setup resumes automatically after you log back in.**

- 3. When asked, pick your LLM provider and paste its API key (input is hidden). This is required — it powers the AI assistant.
- 4. Wait. The **first boot takes about 10 minutes** (up to ~20 on slow disks) while dependencies install and the database is seeded. Every later start takes seconds.
- 5. At the end a green summary prints all URLs and the **superadmin login and password**. Open `http://localhost:3000/backend` and sign in.

B. Completely clean machine (no Git, no repo)

Download just `start-windows.bat` (from the repository's `scripts/windows/` folder on GitHub) and double-click it. It fetches the launcher, installs Git (or falls back to a ZIP download of the repository), clones into `%USERPROFILE%\open-mercato` , and continues as above.

4. What happens, step by step

Step	Expected duration
Preflight (Windows version, virtualization, AppLocker check)	instant
Prerequisites report + install of anything missing (Git, WSL2, Docker/Rancher)	checks instant; installs 5–15 min
Reboot check (auto-resume after login if needed)	instant
Container engine start	30–90 s; first-ever launch up to 5 min
Repository download (standalone mode only)	1–3 min
<code>.env</code> generation (secrets created once, never overwritten)	instant
LLM provider prompt	waits for you
<code>docker compose up</code> (first run builds images)	seconds; first run 10+ min
Database health, application readiness, AI services	first boot ~10 min total; later runs seconds

During long waits the console shows a spinner with elapsed time and a plain-language status. Build progress is also live at `http://localhost:4000` (blank until the dependency install finishes — that is expected).

5. After setup — URLs and daily commands

Service	URL
Admin app	<code>http://localhost:3000/backend</code>
Build/status splash	<code>http://localhost:4000</code>
MCP server health	<code>http://localhost:3001/health</code>

OpenCode agent health	<code>http://localhost:4096/global/health</code>
Keycloak (dev SSO)	<code>http://localhost:8080 (admin/admin)</code>

Action	Command
Start / restart the stack	<code>scripts\windows\start-windows.bat</code>
Stop (data preserved)	<code>scripts\windows\stop-windows.bat</code>
Status of all services	<code>powershell scripts\windows\start-dev.ps1 -Status</code>
Follow application logs	<code>powershell scripts\windows\start-dev.ps1 -Logs</code>
Rebuild the Docker image (after Dockerfile changes)	<code>powershell scripts\windows\start-dev.ps1 -Rebuild</code>
Full reset — DELETES all data	<code>powershell scripts\windows\start-dev.ps1 -Reset</code>
Run without administrator rights	<code>powershell scripts\windows\start-dev.ps1 -NoAdmin</code>
Prefer/install Rancher Desktop instead of Docker Desktop	<code>powershell scripts\windows\start-dev.ps1 -Runtime rancher</code>
Test run with zero changes	<code>powershell scripts\windows\start-dev.ps1 -DryRun</code>

6. Working in your IDE

Edit the project folder (e.g. `C:\Users\<you>\open-mercato`) with any Windows IDE — VS Code, JetBrains, Cursor. Saving a file hot-reloads the app through Docker’s bind mount. **WSL never enters your workflow**: `/mnt/c/...` is only how a WSL shell would see your disk — do not run the project from a WSL terminal, and do not keep a second copy inside a WSL distro.

IntelliSense caveat: `node_modules` lives in a container volume, not in your Windows folder — so TypeScript autocomplete, go-to-definition into dependencies, and ESLint need one of the setups below.

Option	How	When
Attach VS Code to the container (recommended)	Install the “Dev Containers” extension → <i>Dev Containers: Attach to Running Container...</i> → pick the <code>app</code> container → <code>open /app</code>	Full IntelliSense with zero host tooling — the right default on corporate machines
Host <code>node_modules</code> for the editor only	Install Node 24, run <code>corepack enable</code> and <code>yarn install</code> once in the project folder	You prefer a plain host IDE; the app still runs in containers (warnings about optional native modules are harmless)

Full WSL2-native development	Clone inside an Ubuntu WSL distro, IDE via Remote-WSL (see the WSL2 guide in the docs)	Power users wanting maximum filesystem performance; requires a WSL distro and admin
------------------------------	--	---

Tip: “Go to definition” on `@open-mercato/*` imports lands in the monorepo package *sources* — that is expected and usually what you want. After editing a *package* (not the app), also restart the MCP container:

```
docker compose -f docker-compose.fullapp.dev.yml restart mcp.
```

7. Troubleshooting

First rule: the setup transcript is at `%TEMP%\open-mercato-setup\start-dev-*.log`, and container logs are one command away: `docker compose -f docker-compose.fullapp.dev.yml logs -f app` (replace `app` with `mcp`, `opencode`, `postgres` ...).

Installation problems

Symptom	Cause & fix
"App package is not supported..." when installing winget manually	Do not install winget — it is not needed. The launcher downloads official installers directly. Just run <code>start-windows.bat</code> again.
Downloads fail (corporate proxy / air-gapped network)	Either set <code>HTTPS_PROXY</code> for the session, or place the official installers (Git for Windows, Docker Desktop or Rancher Desktop, <code>wsl_update_x64.msi</code>) in an <code>installers</code> folder in the repository root (or point <code>OM_INSTALLERS_DIR</code> at them). The launcher validates and uses them with no network access.
"CPU virtualization appears disabled"	Enable Intel VT-x / AMD-V (SVM) in BIOS/UEFI. On managed machines, ask IT.
"PowerShell is running in ConstrainedLanguage mode"	An AppLocker/WDAC policy restricts PowerShell. Ask IT to allow this script or run it from an exempted path.
No administrator rights at all	Ask IT to install <i>either</i> Docker Desktop (WSL2 backend; add your account to the <code>docker-users</code> group) <i>or</i> Rancher Desktop (dockerd/moby engine). Then run <code>start-dev.ps1 -NoAdmin</code> — everything else works unprivileged (the repo is fetched as ZIP if Git is missing).
UAC prompt cancelled by mistake	Just run <code>start-windows.bat</code> again.

Engine and container problems

Symptom	Cause & fix
"Docker engine did not become ready within 5 minutes"	Open Docker Desktop from the Start menu and finish first-run dialogs (sign-in can be skipped). If it says <i>"WSL 2 installation is incomplete"</i> : run <code>wsl --update</code> in an elevated prompt or install <code>https://aka.ms/wsl2kernel</code> , then re-run.

Both Docker Desktop and Rancher Desktop installed	The launcher uses whichever is already running, else Docker Desktop, and keeps the docker CLI context aligned. Force a choice with <code>-Runtime docker</code> or <code>-Runtime rancher</code> .
Rancher Desktop: <code>docker compose</code> not working	Rancher must run the dockerd (moby) engine (the launcher requests this itself via <code>rdctl</code>). In the Rancher UI: Preferences → Container Engine → dockerd (moby).
"WARNING: DOCKER_INSECURE_NO_IPTABLES_RAW is set"	Benign Docker Desktop networking notice — safe to ignore. It does not stop setup.
"The '<service>' container is crash-looping"	The launcher prints the failing container's last log lines — the real error is there. Most common: <code>.env</code> credentials no longer match an existing database volume → run <code>-Reset</code> (deletes data) or restore the old <code>.env</code> . Stop the restart churn with <code>stop-windows.bat</code> , fix, re-run.
"Port 3000/3001/4096... already in use"	Another process owns the port (the launcher names it). Close it, or override ports in the repo-root <code>.env</code> : <code>APP_PORT</code> , <code>MCP_PORT</code> , <code>OPENCODE_PORT</code> , <code>OM_DEV_SPLASH_PORT</code> , <code>KEYCLOAK_PORT</code> — the launcher follows these automatically.

Application and AI problems

Symptom	Cause & fix
First boot "looks stuck"	It almost never is: the first boot installs dependencies, builds ~20 packages, and seeds the database inside the container (~10 min, up to ~20). Watch <code>http://localhost:4000</code> or the app logs. The launcher fails loudly if something actually breaks.
Browser cannot reach <code>http://localhost:3000</code>	Confirm the app container is running/healthy: <code>-Status</code> . If a firewall/VPN client filters localhost, allow local connections to ports 3000/4000.
AI chat (Ctrl+K) answers with authorization errors / MCP "disconnected"	The MCP key rotates when the database is reset — the OpenCode container reads it only at start. Run: <code>docker compose -f docker-compose.fullapp.dev.yml restart opencode</code> . Verify with <code>curl http://localhost:4096/mcp</code> → should show <code>"status":"connected"</code> .
AI chat gives provider/model errors	Check your provider key in the repo-root <code>.env</code> (e.g. <code>OPENAI_API_KEY</code>) and <code>OM_AI_PROVIDER</code> . After changing them: <code>docker compose -f docker-compose.fullapp.dev.yml restart opencode app</code> .
Login credentials from the summary do not work	If you recreated <code>.env</code> while keeping the old database volume, the printed password may differ from the seeded one (default: <code>password</code>). Cleanest fix: <code>-Reset</code> then a fresh run.
Restarted only the <code>app</code> container and AI tools act strangely	The app rebuild rewrites shared volumes under the MCP server. Restart it too: <code>docker compose -f docker-compose.fullapp.dev.yml restart mcp</code> .

8. FAQ

How long does it take?

First run on a clean machine: ~15–35 minutes total (installers + possible reboot + ~10 min first boot). Every later start: under a minute plus engine startup.

Do I need administrator rights?

Only once, to install Docker/WSL2. If IT pre-installed them, never — use `-NoAdmin`.

We are not allowed to use Docker Desktop (licensing). What now?

Use Rancher Desktop — fully supported. `-Runtime rancher` installs it, or the launcher auto-detects an existing one. It must use the dockerd (moby) engine.

Where are my login credentials?

In the green summary at the end of setup, and in the repo-root `.env` (`OM_INIT_SUPERADMIN_EMAIL` / `OM_INIT_SUPERADMIN_PASSWORD`). They are not written to any log file.

Where does my data live? How do I wipe it?

In named Docker volumes (they survive stops and reboots). Wipe everything with `powershell scripts\windows\start-dev.ps1 -Reset` (asks for typed confirmation).

How do I update to the latest code?

With Git: `git pull`, then `start-windows.bat`. Installed from ZIP: download a fresh ZIP and run the launcher inside it (data lives in Docker volumes, not in the folder).

Can I change the AI provider later?

Yes — edit the repo-root `.env` (provider key + `OM_AI_PROVIDER`, optionally `OM_AI_MODEL`), then restart: `docker compose -f docker-compose.fullapp.dev.yml restart opencode app`.

Which ports are used?

3000 (app), 4000 (build splash), 3001 (MCP), 4096 (OpenCode), 8080 (Keycloak). All overridable in `.env`.

Does it work offline?

Installers can be pre-seeded (the `installers` folder), but the first boot needs the internet for Docker images and npm packages. After the first successful boot, day-to-day work is local.

How do I remove everything?

`-Reset` (removes containers and data volumes), delete the repository folder, and uninstall Docker/Rancher Desktop from Windows Settings if no longer needed.